

Gate E8 report — run_v1.21_r1 (Paper IV v1.21)

Verdict: **PASS with observations.** Evidence: 20_EVIDENCE/E8/.

E8 — Author evidence and packaging revalidation

Author objects examined (manuscript directory, bound by MANUSCRIPT_REVIEW_MANIFEST): `sorry_log_recheck.py`, `SORRY_RECHECK.json`, `RUNNER_SORRY_FIX_NOT_APPLIED.patch`, `AXIOM_COUNT_RECONCILIATION.json`, `EVIDENCE_AND_ANNEX_CORRIGENDUM.md`, `LEAN_IDENTITY_CHECK.json`, `EXTERNAL_EVIDENCE_BINDING.json`, `RESPONSE_TO_AUDIT_v1.21.md`, `EDITORIAL_CHANGE_REPORT_v1.21.md` (grep only). They are not inherited verdicts. Each was contrasted with the auditor's own checks.

Item	Examination	Result
X-28 detector	The pattern was copied literally from <code>sorry_log_recheck.py</code> into the auditor's <code>e8_detector_check.py</code> , not imported. 10 cases were tested, including the real captured negative log (sha256 58441539...). All 1266 existing logs across the 4 sets were scanned.	Correct: all positives detected, all negatives accepted, 0 hits in the logs. The frozen runners are unchanged (sha256 82479893... / 0a7e66a2... equal the manifest), and the patch is proposed and not applied. Mitigated; the historical runners keep their insufficient pattern. OBSERVATION E8-01: the author's rescan omits the 422 logs of author segment 1; the auditor's scan covers them (0 hits). OBSERVATION E8-02: the proposed <code>audit_publication.py</code> patch also adds <code>re.I</code> to the <code>error</code> alternative, which broadens that check (acceptable, but a change of scope).
X-20 axiom count	<code>AXIOM_COUNT_RECONCILIATION.json</code> : 314 + 147 = 461	Equals the auditor's recount on the reused auditor logs (E4_REUSE_CHECK). The erratum "313" of the internal report is corrected by a separate note, without rewriting history. Kept as errata; resolved.
X-25 annex README	The corrigendum reads "§8.2" as "Appendix G.1". The annex ZIP is unchanged (2847a422...).	Resolved by supplement (annex file untouched).
X-19 lakefile residue	Kept deliberately and documented.	Acceptable; OBSERVATION kept.
X-21 / X-29	The corrigendum and §7 record the auditor's byte-identical reproduction and keep the historical doubt.	Correct. The historical limitation remains recorded.
X-22 disagreement	Kept in both the corrigendum and the response.	Correct.
X-26 independence	The corrigendum describes the v1.2 run as a separate fresh session.	Correct for run_v1.2_r1. This v1.21 revalidation ran in the same session as run_v1.2_r1 , as the auditor declares (not blind, not independent of that run).
AUD-P1, C-01...C-08	Preserved in run_v1.2_r1 and unaltered (E0: prior manifest unchanged).	Correct.
Evidence binding	<code>EXTERNAL_EVIDENCE_BINDING.json</code> : 18 items and the archive; all hashes match.	Correct.
Response claims	"No theorem, hypothesis, constant, equation tag or Lean block changed; five declared renamings".	Confirmed by the auditor's own diff (E2). However, the author's header comparison "allowing only monospacing" let a descriptive heading word become a code identifier (NEW-01, E1/E6).
Manuscript package	MANUSCRIPT_REVIEW_MANIFEST has 88 files; the r1 ZIP has 90 members (88 + manifest + its sidecar), all hashes equal. The non-r1 <code>PAPER_IV_v1.21_REVIEW_PACKAGE.zip</code> also exists in the directory but is not the bound target.	Correct. OBSERVATION E8-03: an earlier unbound package sits next to the bound r1 package.

E8 verdict: PASS with observations.

PDF generated by pandoc (HTML) + PyMuPDF Story from the Markdown source; not compiled from TeX. The TeX file alongside is pandoc's standalone conversion of the same Markdown.